Lab 7 - PN Junction Simulations

Circuits, Devices, and Transduction

ES 152 - Fall 2020

Objectives

1. Perform simulations of PN junctions
2. Gain intuition for the relationship between the physical properties and operating characteristics of a PN junction

Introduction

A PN junction is one of the most fundamental semiconductor devices—in fact, it forms the basis of many other devices, such as the bipolar junction transistor (BJT) and light emitting diodes (LED). As the name suggests, a **PN** junction is created by forming a junction between a p-type and an n-type semiconductor. The material properties, physical structural characteristics, and external environmental conditions determine the operating characteristics of the PN junction. In this lab, you will explore, through numerical simulations, how these parameters (structural, material, and environmental) control the operating characteristics of a PN junction.

We will be using an online computational platform for performing the simulations called nanoHUB. nanoHUB is a collection of computational tools for nanotechnology created by researchers at various universities. The underlying computational engine for these PN-junction simulations is called PADRE, which in the simplest form uses the drift-diffusion model to calculate the current-voltage characteristics. We will not be directly interacting with PADRE, which is a command-line tool, but will be using a web-interface specifically designed to run PN-junction simulations. Together, these web-based tools are called the PN-junction Lab. If you want to know more about the theory and computational method used by this tool, go to the Appendix, otherwise continue reading.

Accessing the PN-junction Lab Tool

Before being able to use the PN-junction Lab, you will have to register for an account with nanoHUB. Click on this link to register for an account. This should take no more than 2 minutes. Once you have activated your account, click on this link to go to the PN-junction Lab tool. Make sure you are logged into your nanoHUB account. Then click on the *Launch Tool* button to get started (see Figure 1).

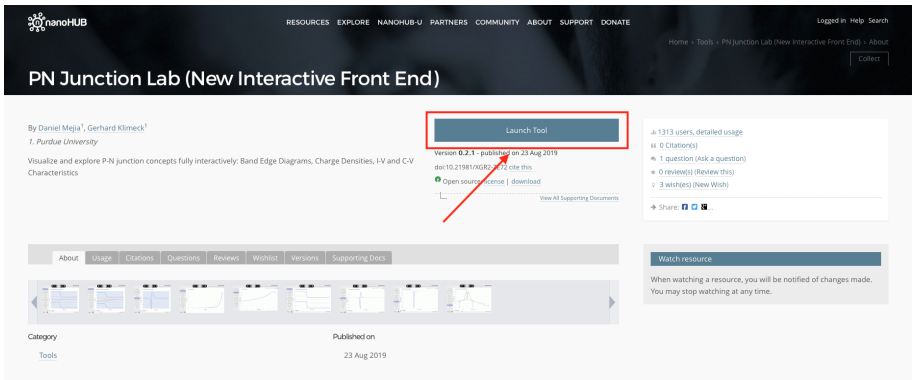

Figure 1: Launching the PN-junction tool. Note that you need to be logged into your account to be able to launch the tool.

Getting Familiar with the Interface

Let's take a closer look at the interface for the PN-junction Lab tool. When you launch the tool, it can take a few seconds to a minute to create and start your session (it's running a default simulation), so hold on. Once the session is created, you will see a screen as shown in Figure 2.

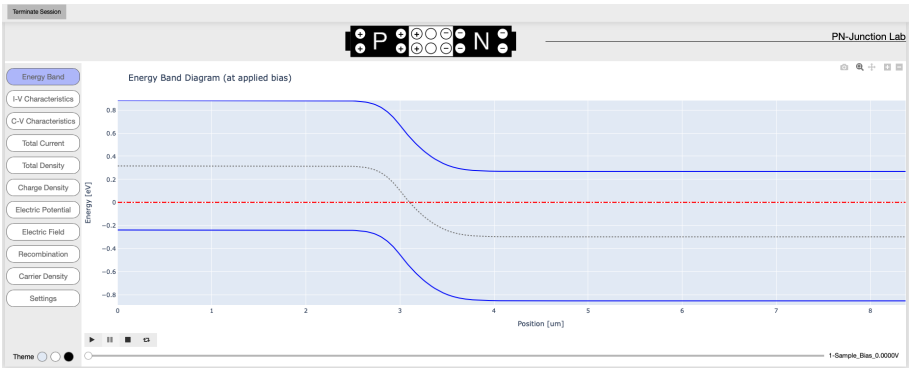

Figure 2: The screen you will see upon launching the PN-junction tool. Note that it can take a few seconds to a minute to create and start a session.

On the left side of this page are various buttons for viewing plots of different results, such as *Energy Band*, *IV Characteristics*, et cetera. Some of the plots have a little bit of interactivity where you can play through the plots for different applied bias. The last button on the left side, called *Settings*, is where you will input your trial-specific configurations. Here, you will be able to change the geometry as well the material and physical characteristics (doping density, for instance) of your PN-junction in the simulation. If you make any changes in the *Settings* screen and go back to any of the plot screens, the simulation will automatically re-run and update the results after it finishes

computing (this might take a few moments). On the top-right side of a plot are buttons to take a screenshot, zoom, pan, and reset. Finally, there is a button on the top-left side of the page for terminating the session. Make sure you have downloaded anything you need to save before terminating your session. If you terminate your session, the next time you launch one, it will start again from default—your work will not be saved. In Figure 3, all of these interface elements are highlighted.

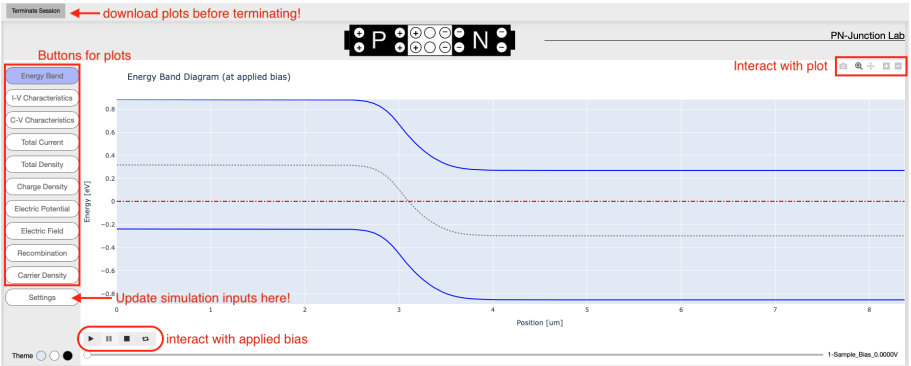

Figure 3: Main interface elements in the PN-junction tool. Note that all the inputs are under the *Settings* button.

Now, let's look at the *Settings*. This is where all the inputs for the simulation are. Besides properties of the device, there are a few settings that control certain numerical aspects of the computation and are useful to understand. It is instructive to take a look at the *Settings* for the default model, which is what we will be discussing below. The results you see on the plots when you launch a session are matched to this default model.

The first thing to notice is that the inputs are broken down into three categories named *Structure*, *Materials*, and *Environment*, all of which are arranged in their respective tabs. The *Structure* tab contains the inputs for designing the geometric structure of the device and are mostly self-explanatory; for example, the length of the p, n, and i-type regions (i refers to intrinsic), acceptor and donor impurity concentrations. You will also notice inputs for p, i, and n-type nodes. This input refers to the level of discretization (or resolution) of the regions. In more literal terms, this is the number of points at which the underlying coupled partial differential equations are solved. The larger you make this number, the higher the resolution and more accurate the solution will be (***but*** it also lengthens the compute time). So, you'll want to strike a balance. The default model uses 20 nodes per micron, which is a good baseline to keep in mind. Of course, it's just a baseline. If—for example—you uses a large doping density, then you might have to increase the number of nodes for the solution to converge.

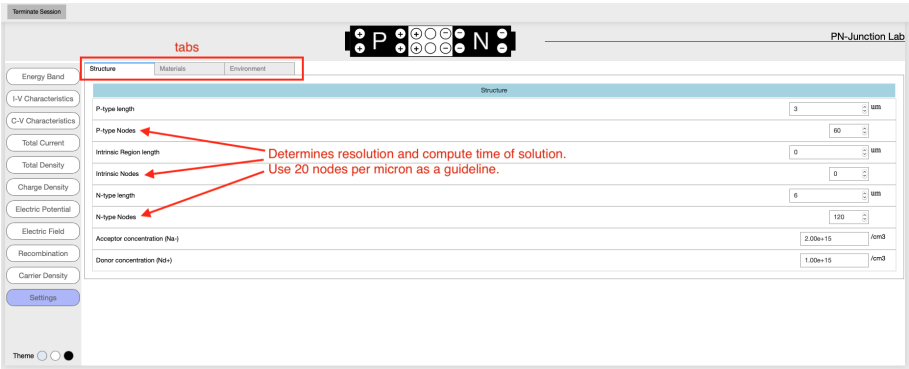

Figure 4: Input options in the *Structure* tab of the *Settings* button. The number of nodes should be a balance between resolution for desired accuracy and compute time.

The second tab, *Materials*, is where you can choose your semiconductor material for your PN-junction. (You can pick from Si, Ge, GaAs, and InP. Which of these would be useful for an LED and why?[1]) For a given material you also can input the minority carrier lifetimes. These are the average amounts of time an electron spends in a p-type region before recombining with a hole (and *vice versa*). These numbers vary with material and for a given material this varies with impurity density. The value for the default model is a bit on the shorter side. A typical value is the range of a few hundred nanoseconds to a few microseconds.

Let's try something out. Change the minority carrier lifetimes for both holes and electrons to be 1 microsecond (1×10^{-6} s) and click on the *I-V Characteristics* plot button on the left. First you will notice that the simulation will recalculate the solution and you will see the message `waiting results from nanoHUB`. This can take a minute to completion. Now, in this I-V plot (current *v.* voltage), you will notice a *Compare* button in the lower-left side of the screen.

Clicking on this button allows you to see the results from the previous simulations in your current session. Unfortunately, all the previous simulations are shown in the same shade of gray, so this is useful only for comparing against one previous solution. If you clicked on *Compare*, it will be replaced with a *Clear History* button. Clicking on this will remove all previous simulation results. So, be careful if you want to compare results. If you switch to any different plot screen, the plots are reset and the *Compare* button will come back. If you clicked on the *Compare* button for the *I-V Characteristics* plot after changing the minority carrier lifetimes as mentioned above, you will see the result shown in Figure 5

[1] An LED is essentially a PN-junction (with some optimizing modifications) that produces light when the electrons and holes recombine. In order to produce light you need a material that has a direct bandgap, for example, GaAs and InP, wherein an electron can *directly* emit a photon. Check out section 4.7.4 of Sedra and Smith for more.

Figure 5: Comparing I-V characteristics for the default PN-junction model and updated model with minority carrier lifetimes changed to 1 microsecond. Note the *Clear History* button at the lower-left side of the screen will remove all previous simulation results from the cached memory in your current session.

Notice that the **shorter minority carrier lifetime** results in **lower current density**. Why is that so?[2]

Finally, the last tab in the *Settings* screen has inputs related to environmental factors, such as temperature and applied voltage. There is a third input called *Number of points* which controls how many test points you have in the applied voltage, which will be selectable within the range from 0 V to the value you enter in *Applied Voltage*. With this, you should have all the necessary information to run simulations in the PN-junction tool and answer the questions in the our accompanying Python notebook. If you'd like more background information about semiconductors and PN junctions, please consider reading through Chapters 3 and 4 in Sedra and Smith. Consider Sections 3.4 through 3.6 and Sections 4.1 through 4.4. ES173 is a great follow-on course to learn more, too!

Outline

1. Create login credentials and explore the nanoHUB PN Juntion Lab tool
2. Validate numerical expression for built-in voltage and depletion region width
3. Explore PN Junction characteristics for forward and reverse bias conditions
4. Design a Si PN-junction diode with a built-in potential near $0.7V$ with strict current density minimum given a forward bias and leakage maximum given a reverse bias
5. Complete the Python notebook template

[2]Minority carrier lifetime is directly proportional to diffusion length. So, the longer the lifetime of say electrons in the p-region, the longer their diffusion length, which means that the exponential decay of electron density in the p-region is more gradual and hence $\frac{dn_p}{dx}$ is smaller (*i.e.*, the concentration gradient of minority carriers), making the diffusion current smaller. See Section 3.5.2 of Sedra and Smith for more.

Report

The only deliverable for this lab is a report generated from the provided Python notebook template. This template will guide your theoretical and experimental exploration, and we ask you to capture relevant information. As a report, the notebook should include your observations, perspectives, and conclusions.

Naming convention: es152_ lab7_ lastname.pdf

Appendix

A summary of the computational method behind the PN-junction tool is provided. This tool uses the well-known drift-diffusion model in 1 dimension to calculate the operating characteristics of a PN-junction. The unknowns for which we are trying to solve are the current density, $J = J_n + J_p$, the carrier density, n and p, and the electric field, E. The letter n denotes electrons while p denotes holes (quasi-particles representing the lack of an electron—the "hole" left by an electron). As you can see, there are five unknowns (J_n, J_p, n, p, E), which are determined by solving the set of equations listed below.

Drift-diffusion equations for current density

$$J_n = qn\mu_n E + qD_n \frac{\mathrm{d}n}{\mathrm{d}x} \tag{1a}$$

$$J_p = qp\mu_p E + qD_p \frac{\mathrm{d}p}{\mathrm{d}x} \tag{1b}$$

Here, J_n and J_p denote the current density while n and p are the charge carrier densities from electrons and holes, respectively. q is a unit of charge and E is the electric field. μ and D represent the carrier mobility and the diffusion coefficient, respectively. The first term on the right-hand side of the above equation represents drift current, *i.e.*, current produced via carriers movement (due to an electric field). The second term on the right-hand side demotes diffusion current, *i.e.,* the current generated via carriers spreading (due to a concentration or density gradient).

Poisson's equation for electrostatic field

$$\epsilon \frac{\mathrm{d}E}{\mathrm{d}x} = q(p - n + N_d^+ - N_a^-) \tag{2}$$

Here, N_d^+ and N_a^- denote the density of ionized donor and acceptor impurities, respectively. ϵ is the dielectric constant of the semiconductor material. You might have seen this equation in some other forms such as, $\epsilon \nabla^2 \phi = -\rho$, where $\nabla \phi = E$ is the electric potential and ρ is the charge density. (To be more pedantic, the above equation is actually more commonly referred to as Gauss's Law, while the one with the electric potential in the previous line is referred as Poisson's equation for electrostatics.)

Continuity equation for conservation of charge

$$\frac{\mathrm{d}n}{\mathrm{d}t} = \frac{1}{q}\frac{\mathrm{d}J_n}{\mathrm{d}x} + U_n \tag{3a}$$

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{1}{q}\frac{\mathrm{d}J_p}{\mathrm{d}x} + U_p \tag{3b}$$

Here, U_n and U_p represent the net rate of generation/recombination of electrons and holes, respectively. This equation basically states that the rate of change of carrier density (the left hand side) is equal to the current entering or leaving and the net change in carriers due to generation or recombination.

The drift-diffusion equations (1) determine the current densities but solving this equation requires spatial knowledge of the electric field (E) as well as the charge carrier densities (n, p). Poisson's equation (2) can be used to calculate the electric field given knowledge of charge densities. Finally, the continuity equation (3) which essentially describes the conservation of charge principle adds the additional relationship needed to solve for the charge densities. So together, by solving these equations iteratively, one after another starting from an initial guess, we can obtain a converged solution of all the unknowns.

Computationally, these equations are solved by discretizing the variables on a mesh using finite-differences and iterating until the solution converges, which is determined by keeping track of the difference between the last two consecutive solutions. There are numerical constraints on the mesh size and the time step for a stable convergence of the solution. These constraints are related to the Debye length (*i.e.*, the effective "size" of an electron in a sea of other electrons and charges) and the dielectric relaxation time (*i.e.*, the characteristic time it takes for electron density fluctuations to decay).